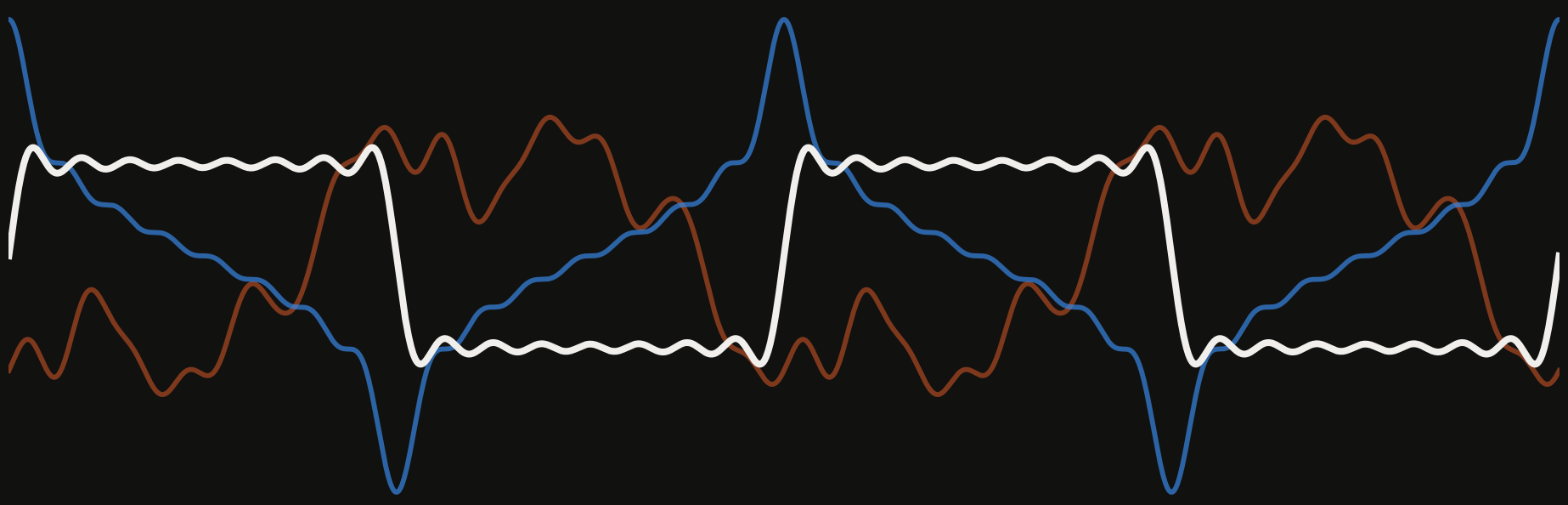
The half of the FFT your analyser throws away

Where did the shape go?

Real and imaginary parts, the mirror, and why phase is the shape.



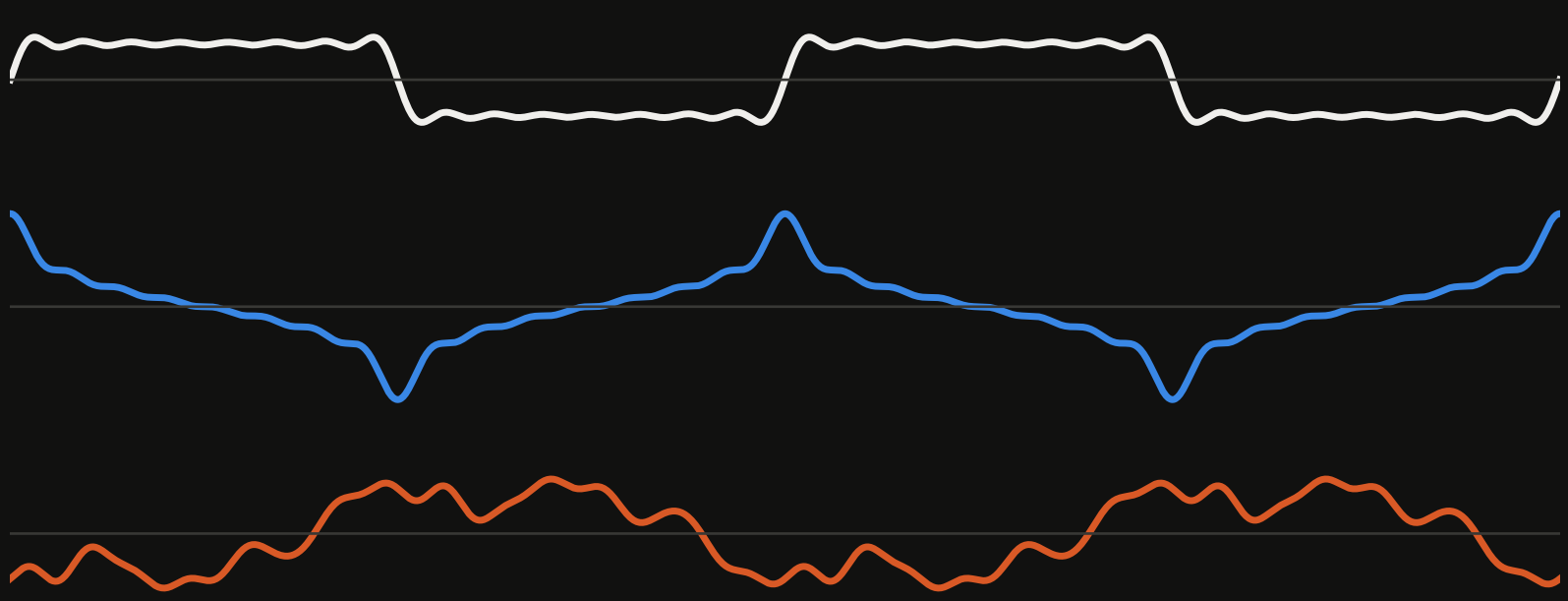
FFT series · Part 1

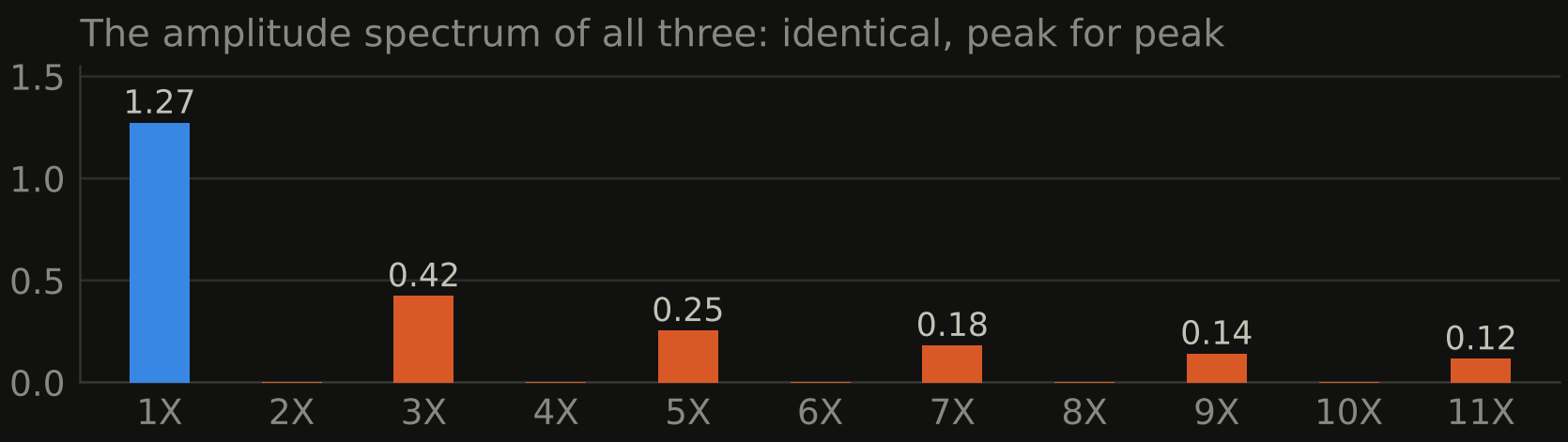
Swipe →

THE PUZZLE

Three signals, one spectrum

A square wave, a sharp spike and something that looks like noise. Run all three through an FFT:





Same harmonics, same heights, three different shapes. So the spectrum can't be the whole story.

WHAT THE FFT RETURNS

Every bin is two numbers

An FFT doesn't return heights. For each frequency bin it returns a complex number: a real part and an imaginary part.

$$X_n = \frac{1}{N} \sum_{k=0}^{N-1} x_k \, e^{-i 2\pi n k / N}$$

Eight samples of $x(t) = 0.6 \cos \omega_0 t + 0.8 \sin \omega_0 t + 0.5 \cos 2\omega_0 t$ give:

bin	real	imaginary	size $\lvert X_n \rvert$
X0	+0.00	+0.00	0.00
X1	+0.30	-0.40	0.50
X2	+0.25	+0.00	0.25
X3	+0.00	+0.00	0.00
X4	+0.00	+0.00	0.00
X5	+0.00	+0.00	0.00
X6	+0.25	+0.00	0.25
X7	+0.30	+0.40	0.50

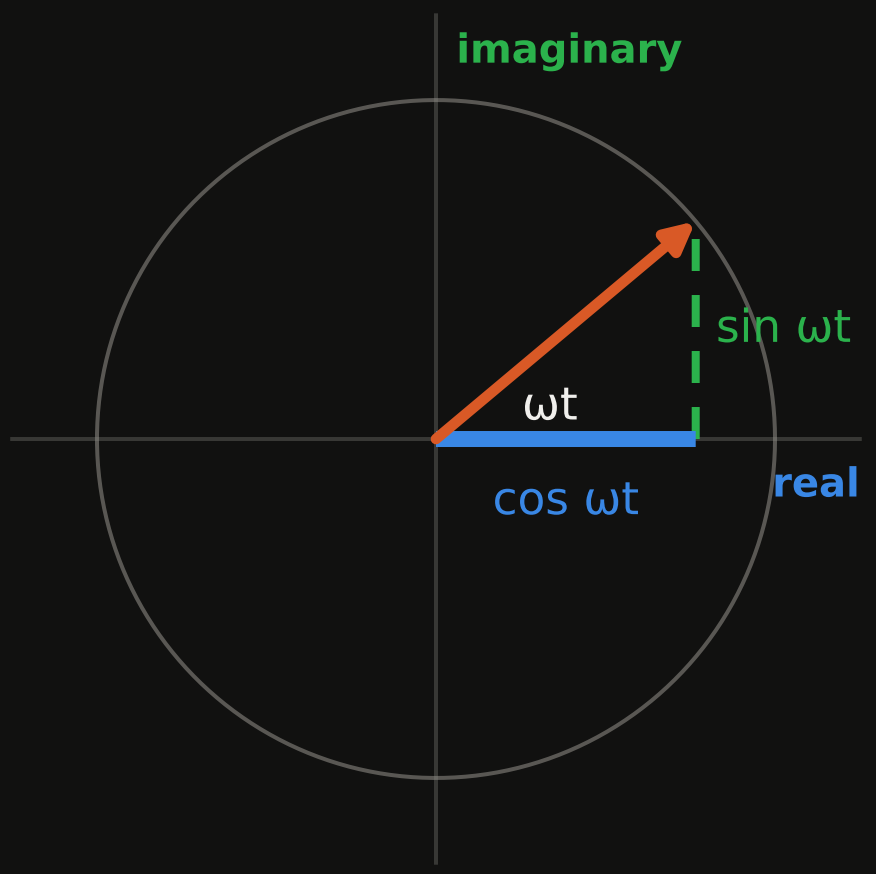
A spectrum plot shows only the size column (doubled, slide 5). The other two columns are where the shape lives.

EULER’S FORMULA

Each frequency is an arrow

Think of one frequency as an arrow turning round a circle. Its shadow along the real axis is a cosine; its shadow up the imaginary axis is a sine.

$$e^{i\omega t} = \cos \omega t + i \sin \omega t$$



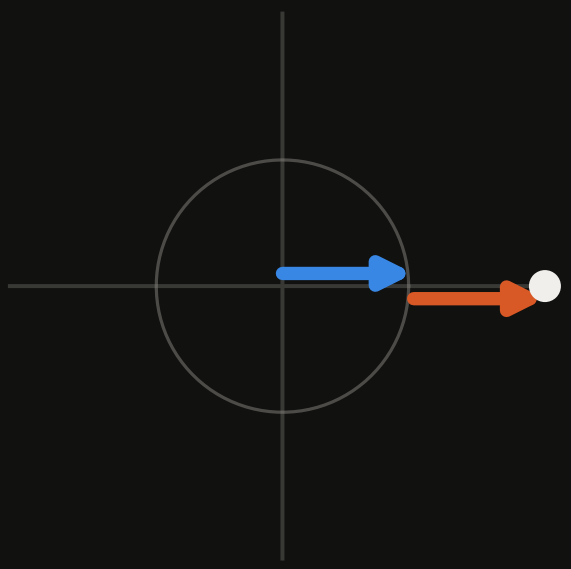
One turning arrow, two shadows: a cosine on the real axis and a sine on the imaginary axis.

TWO ARROWS

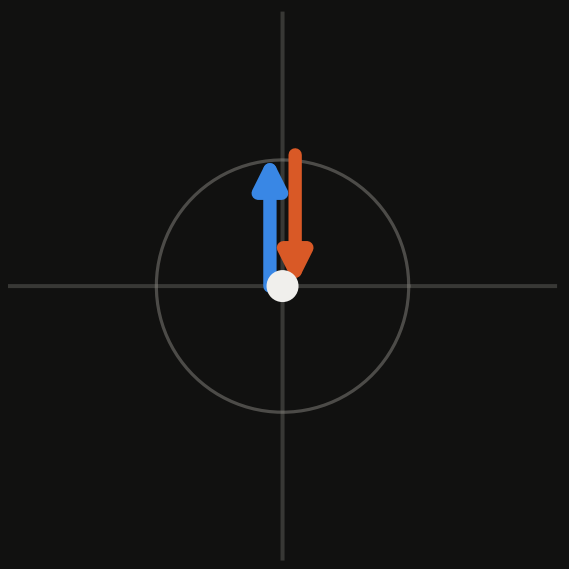
Why half the FFT is a mirror

A real vibration moves along one line. One spinning arrow can’t: its tip also goes up and down. Add a twin spinning the other way, and the ups and downs always cancel:

$$\cos \omega t = \tfrac{1}{2}\, e^{\,i\omega t} + \tfrac{1}{2}\, e^{\,-i\omega t}$$



on top of each other:
one arrow, twice as long



a quarter turn later:
one up, one down, zero

So every peak appears twice, at +f and −f, each half the size. For a real signal the second half of the FFT is the mirror image (complex conjugate) of the first:

$$X_{N-n} = \overline{X_n}, \qquad \text{peak} = 2\,|X_n|$$

REAL AND IMAGINARY

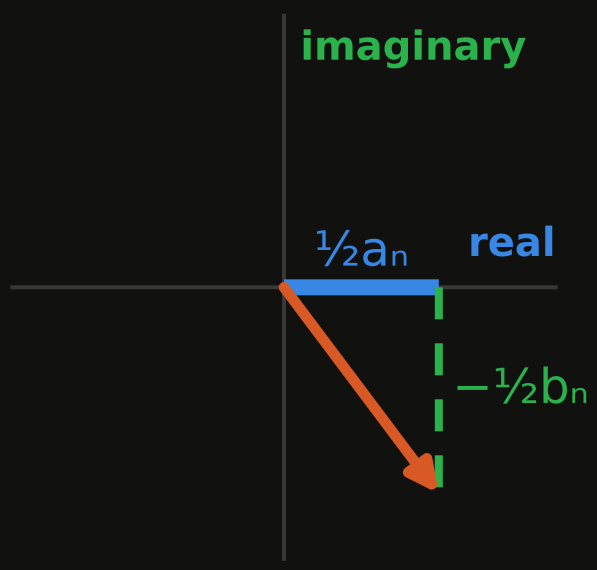
What the two numbers measure

Write a harmonic the usual way, as a cosine part and a sine part:

$$a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)$$

Put Euler’s formula into the FFT and it splits into two sums. Compared with the Fourier coefficients:

$$X_n = \tfrac{1}{2}\,(a_n - i\,b_n)$$



Real part = half the cosine amount.
Imaginary part = minus half the sine amount.

LENGTH AND ANGLE

Size and phase

Write the same harmonic as one cosine with a size and a phase, and match it to the arrow:

$$a_n \cos n\omega_0 t + b_n \sin n\omega_0 t = c_n \cos(n\omega_0 t + \theta_n)$$

$$c_n = \sqrt{a_n^2 + b_n^2} = 2\,|X_n|$$

$$\theta_n = \arg X_n, \qquad \tan\theta_n = \frac{\operatorname{Im} X_n}{\operatorname{Re} X_n}$$

The arrow’s length is the peak on your spectrum. Its angle is the phase: when in each turn that harmonic peaks.

> **Watch the quadrant. If the real part is negative, a calculator’s $\tan^{-1}$ is 180° out. Use the signs of both parts (atan2).**

WORKED EXAMPLE

One bin, by hand

Same signal as slide 3, eight samples at 45° steps. For bin 1, multiply each sample by cos θₖ and sin θₖ and add:

$$\sum x_k \cos\theta_k = 2.4, \qquad \sum x_k \sin\theta_k = 3.2$$

$$X_1 = \frac{2.4}{8} - i\,\frac{3.2}{8} = 0.3 - 0.4\,i$$

$$2|X_1| = 2\sqrt{0.3^2 + 0.4^2} = 1.0$$

$$\theta_1 = \tan^{-1}\left(\frac{-0.4}{0.3}\right) = -53.1\,^\circ$$

Check: ½(a₁ − i b₁) = ½(0.6 − 0.8i) = 0.3 − 0.4i ✓

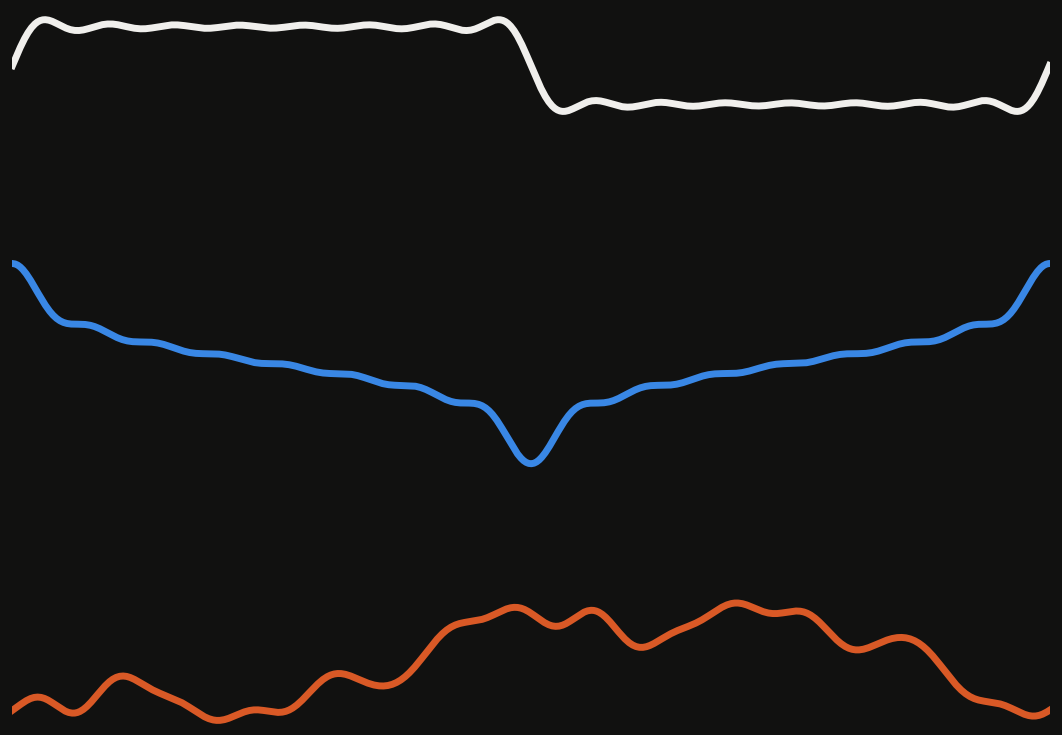
> **So the 1X part is 1.0·cos(ω₀t − 53.1°). The spectrum keeps the 1.0. The phase keeps the −53.1°.**

Every line of this, step by step, with your own numbers:
nodallab.dev → Lab → An FFT by hand.

BACK TO THE PUZZLE

Phase is the shape

The three signals from slide 2 share every size.
Only the phases differ:



all phases 0: the sines start together

all shifted 90°: every peak lines up

random phases: no pattern

$$|X_n| \text{ fixed}, \qquad \arg X_n \text{ changed} \Rightarrow \text{a different waveform}$$

The amplitude spectrum keeps the sizes and drops the timing. Phase is what tells a spike from noise.

MEASURING IT

Phase needs a reference

Shift the same signal later in time by τ, and every bin turns:

$$x(t-\tau) \;\Rightarrow\; X_n\, e^{-i\, n\omega_0 \tau}$$

The sizes don’t change, but each phase moves by $n\omega_0\tau$: twice as far at 2X as at 1X. So a phase on its own means nothing. It only means something against a fixed reference:

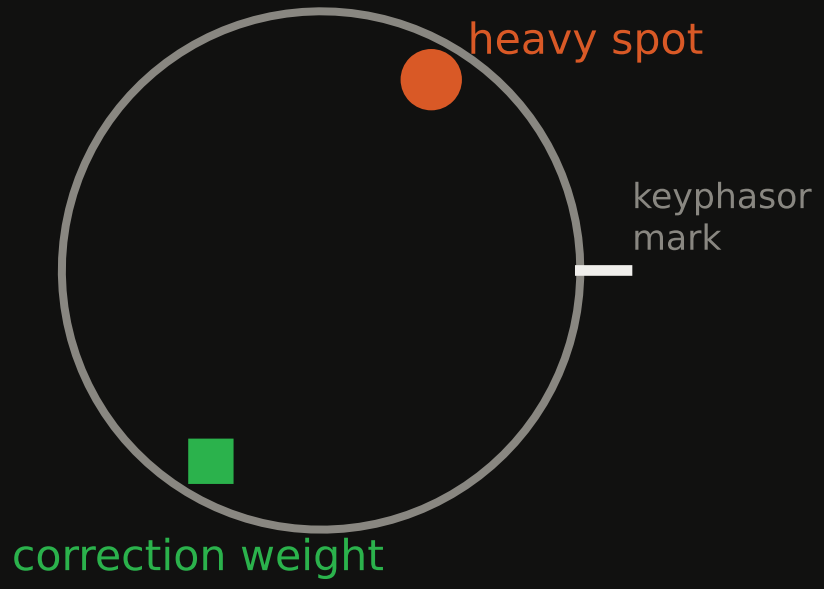
- Absolute phase: a once-per-revolution mark (tachometer or keyphasor) starts the clock at the same shaft angle every time.
- Relative phase: two sensors measured at the same moment. The record start cancels in the difference.

> **Read the phase difference between points, or against the shaft, never a single phase on its own.**

WHERE THE HEAVY SPOT IS

In the field: balancing

An unbalanced rotor vibrates at 1X. Measured against the keyphasor, the 1X phase tells you where, round the rotor, the heavy spot is, allowing for the lag between the force and the response.



No phase, no balancing. With a trial weight to calibrate the rotor, the size sets how much correction and the phase sets where it goes.

MISALIGNMENT

In the field: across a coupling

Measure 1X on both sides of a coupling at the same time and compare the phases.



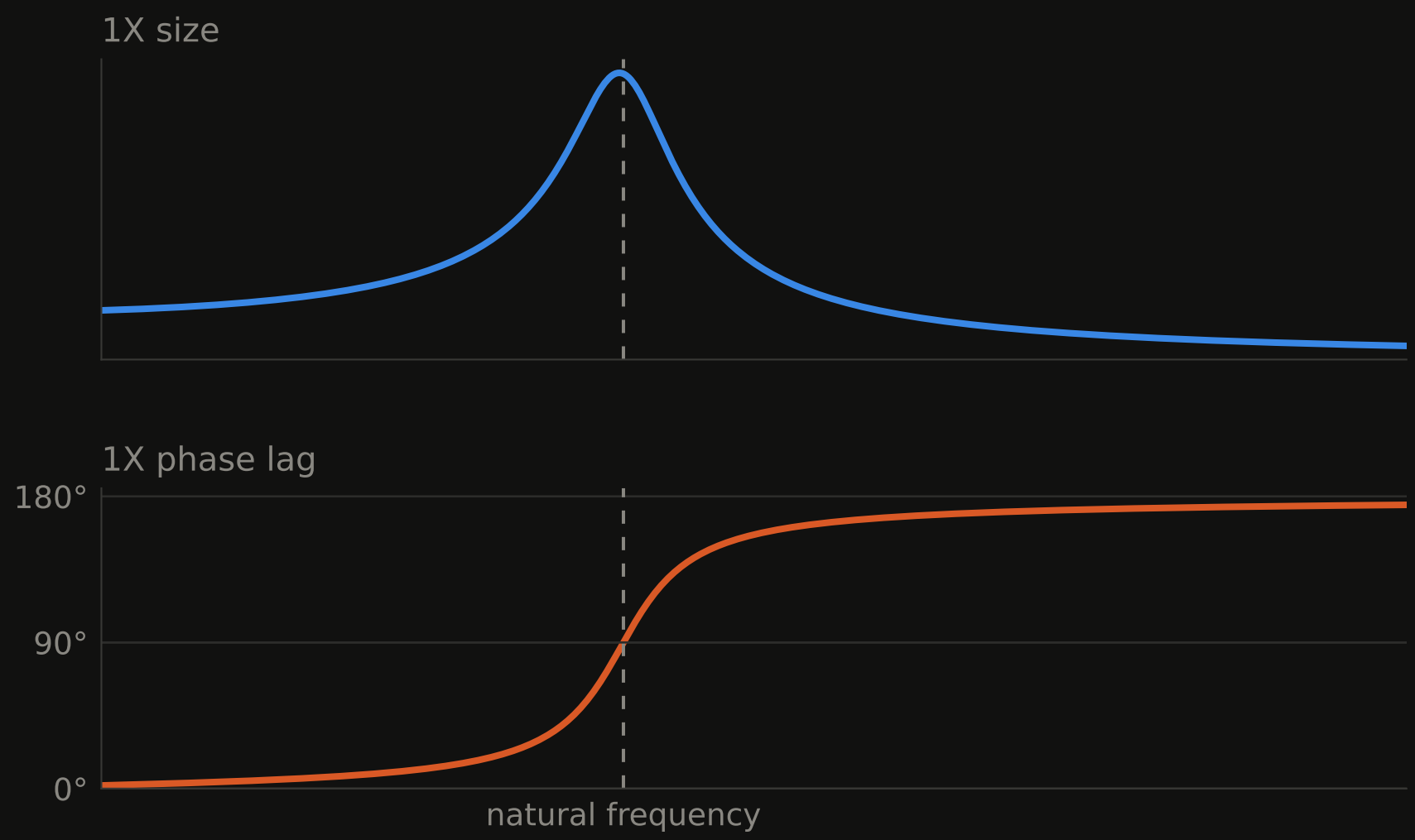
A phase shift of about 180° across the coupling, especially in the axial direction, is a classic sign of misalignment. Read it together with the amplitudes and directions, not on its own.

> **Why phase can tell imbalance from misalignment is a story for another post.**

THE 90° CLUE

In the field: through a resonance

Run a machine up through a natural frequency and plot 1X size and phase against speed (a Bode plot):



The phase swings by about 180°, passing 90° at the natural frequency, close to the amplitude peak.

> **A peak with a big phase swing is a resonance. A peak without one probably isn’t.**

TAKEAWAYS

What this means for the analyst

Every bin is two numbers.

Real: half the cosine part. Imaginary: minus half the sine part.

The top half is a mirror.

Real signals give X_n and its conjugate. Analysers drop one and double the other.

Size is the peak, angle is the phase.

$2|X_n|$ is what you see. $\arg X_n$ is what usually gets thrown away.

Phase is the shape.

Same sizes, different phases, different waveform.

Phase needs a reference.

A keyphasor or a second sensor. Compare phases, don’t read one alone.

Size tells you how much. Phase tells you where and why.

REFERENCES

Sources & further reading

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The Fourier Transform and Its Applications. McGraw-Hill. (Shift theorem: a time delay rotates the phase.)

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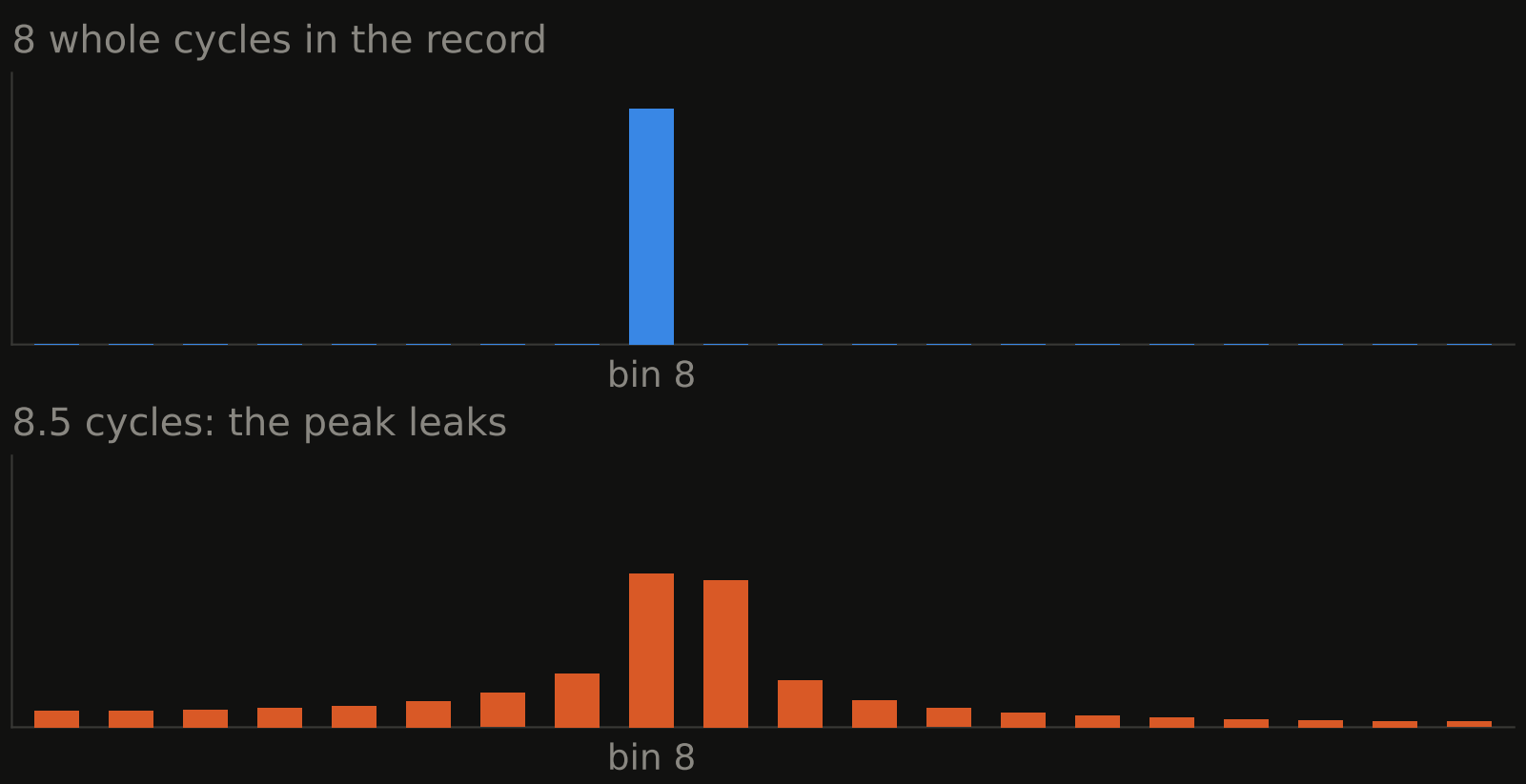
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COMING SATURDAY

Next: when the signal doesn’t fit the window

Everything here assumed the record holds a whole number of turns. When it doesn’t, one clean peak smears across its neighbours. That’s leakage.



See you Saturday!